



# PLA Production & Development of Bioplastics

By Sagar Naik  
2GP25CS403

# **Agenda**

- What are Bioplastics? (The Big Picture)
- Introduction to PLA (Polylactic Acid)
- Step-by-Step PLA Production Process
- Key Developments in Bioplastic Technology
- Current Applications & Market Trends
- Challenges & Future Outlook

# What are Bioplastics?

- **Definition:** A family of materials that are either:
  - **Bio-based** (derived from biomass, e.g., corn, sugarcane)
  - **Biodegradable** (can break down via microorganisms)
  - **Both** (e.g., PLA) or **Neither** (e.g., bio-based PET)
- **Key Distinction:**
  - Bio-based  $\neq$  Always biodegradable (and vice versa)
- **Main Families:**
  - Starch blends
  - PLA (Polylactic acid)
  - PHA (Polyhydroxyalkanoates)
  - Bio-PE / Bio-PET (drop-in replacements)

# **Why Develop Bioplastics? Drivers of Innovation**

## **Environmental:**

- Reduce fossil fuel dependence.
- Lower carbon footprint (PLA emits ~80% less CO<sub>2</sub> than traditional plastics).
- Address plastic pollution & microplastics.

## **Economic:**

- Volatile oil prices.
- Growing consumer demand for sustainable products.

## **Regulatory:**

- Plastic bans in 120+ countries.

# **Spotlight on PLA – The Leading Bioplastic**

- **Full Name:**
  - Polylactic Acid (or Polylactide).
- **Rank:**
  - Most produced bioplastic by volume (excluding starch blends).
- **Key Properties:**
  - Biodegradable (industrially compostable).
  - Renewable origin (~100% bio-based).
  - High transparency & gloss (like PET).
  - Good barrier against fats & oils.
- **Key Weakness:**
  - Poor heat resistance & brittleness (main development targets).

# PLA Production

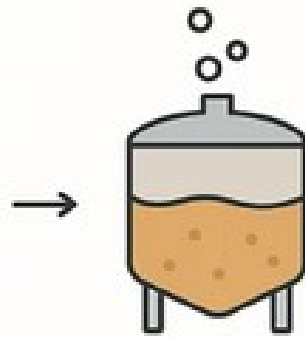
RAW MATERIAL  
SOURCING AND  
STARCH EXTRACTION



Starch  
Extracted

Wet milling

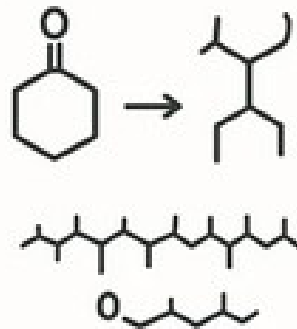
FERMENTATION  
AND LACTIC ACID  
PRODUCTION



Lactic Acid

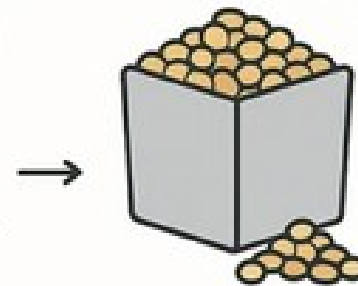
Fermentation

POLYMERIZATION  
INTO PLA



PLA

PROCESSING  
INTO PELLETS



Pellets

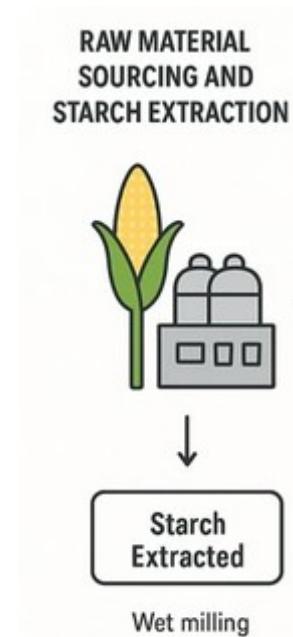
# Step 1: Feedstock & Sugar Extraction

- **Raw Materials:**

- Corn starch (USA – primary)
- Sugarcane / Cassava (Asia, Brazil)
- *Future:* Agricultural waste (non-food biomass)

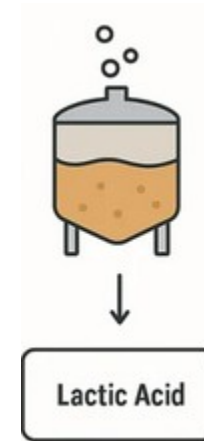
- **Process:**

- Milling to separate starch.
- Enzymatic hydrolysis (enzymes like amylase break starch into glucose/dextrose).



## Step 2: Fermentation (Glucose → Lactic Acid)

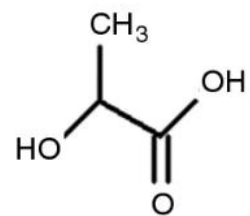
- **Process:**
  - Similar to brewing beer or making yogurt.
- **Microorganism:**
  - *Lactobacillus* species (or genetically engineered microbes).
- **Reaction:**
  - Glucose → Lactic Acid + Water + CO<sub>2</sub>
- **Conditions:**
  - Anaerobic (no oxygen), pH ~5-6, temperature ~45°C.
- **Output:**
  - Crude lactic acid broth (~85-95% purity).



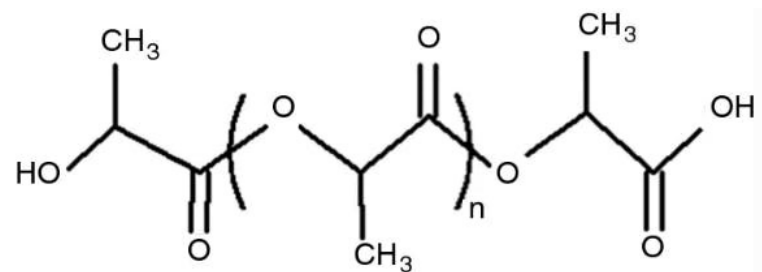
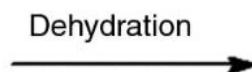


# **Step 3: Purification & Polymerization**

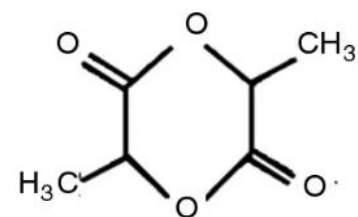
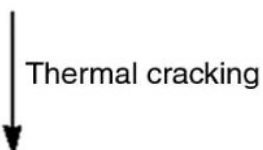
- **Purification:**
  - Distillation, filtration, ion-exchange → >99.5% pure L-lactic acid.
- **Polymerization (The key chemical step):**
  - Lactic acid molecules are heated to form a low-weight oligomer.
  - Oligomer is converted into **Lactide** (a cyclic dimer).
- **Ring-Opening Polymerization (ROP):**
  - Lactide is opened and linked into long PLA chains.
- **Catalyst:**
  - Traditionally tin octoate  
(being replaced by safer, food-grade catalysts).



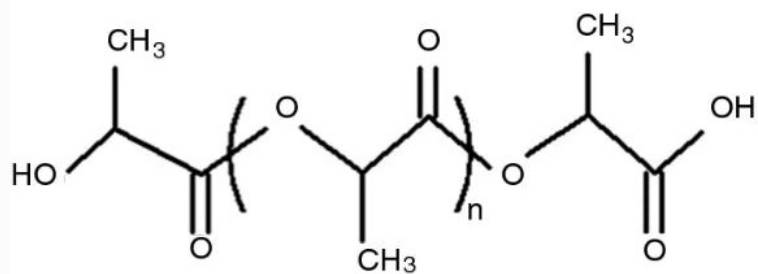
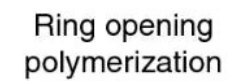
LACTIC ACID



LACTIC ACID OLIGOMERS



LACTIDE



PLA

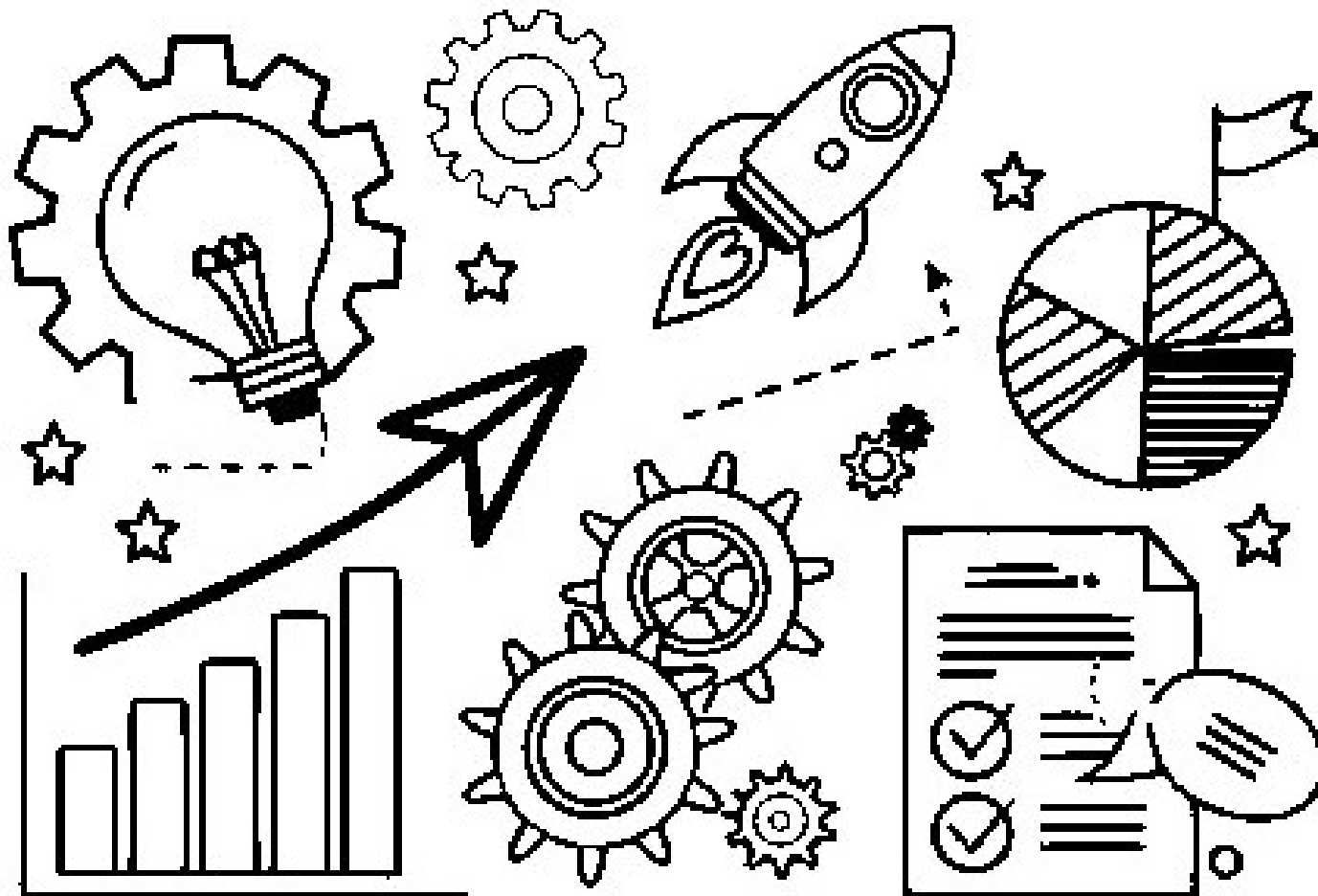


# The Isomer Game

## Tuning PLA Properties

- PLA has two optical isomers:
  - **PLLA** (L-lactic acid) and **PDLA** (D-lactic acid).
- **High PLLA (>99%):**
  - Semi-crystalline
  - Higher melting point (~170°C)
  - Stronger, more rigid
- **Low PLLA / High PDLA:**
  - Amorphous (clearer)
  - Lower melting point (~130°C)
  - Better for films
- **Stereocomplex PLA (PLLA + PDLA blend):**
  - Melting point ~230°C (breakthrough for heat resistance!)

# Development



# #1 – Improving Heat Resistance (The "Hot Coffee" Problem)

- **Problem:**
  - Standard PLA softens at  $\sim 55^{\circ}\text{C}$  (deforms in hot liquid).
- **Development Solutions:**
  - **Annealing:** Heat treatment to increase crystallinity.
  - **Stereocomplex PLA:** Blending PLLA & PDLA (melts at  $\sim 230^{\circ}\text{C}$ ).
  - **Nucleating agents:** Add talc, LAK, or other minerals to speed crystal formation.
  - **Result:** PLA that can withstand microwave & dishwasher conditions.



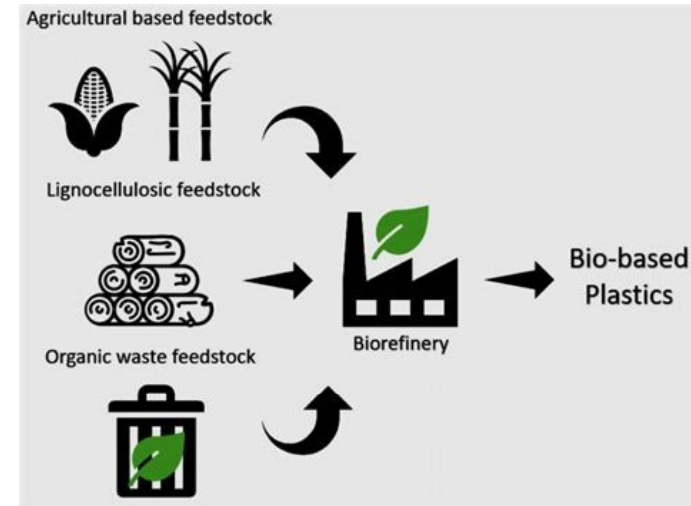
## **#2 – Making PLA Tough (Not Brittle)**

- **Problem:**
  - PLA has low impact strength (brittle – like glass).
- **Development Solutions:**
  - **Plasticizers:** Add glycerol, citrate esters (increase flexibility).
  - **Blending:** Mix with other biopolymers like PBAT (flexible) or PCL.
  - **Copolymerization:** Insert softer segments into the PLA backbone.
  - **Nanocomposites:** Add nano-cellulose or nano-clay.
  - **Result:** PLA films for bags, flexible packaging, and textiles.

# #3 – From Food Crops to Waste

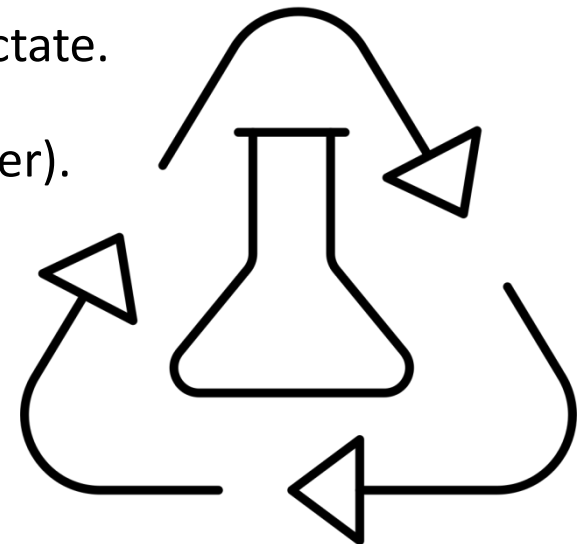
## Feedstock

- **Problem:**
  - Using corn/sugarcane competes with food supply & land use.
- **Feedstock:**
  - Lignocellulosic biomass  
corn stover, wheat straw, wood chips.
- **Challenge:**
  - Breaking down lignin (the tough plant cell wall).
- **Goal:**
  - Low-cost, non-food, waste-based PLA.



# #4 – End-of-Life & Chemical Recycling

- **Current End-of-Life:**
  - Industrial composting (58°C, high humidity) → CO<sub>2</sub> + H<sub>2</sub>O.
- **Problem:**
  - Most PLA ends in landfill or incineration due to lack of composting facilities.
- **Development: Chemical Recycling (Closed Loop)**
  - **Hydrolysis:** PLA + water + heat → Lactic acid (can be re-polymerized).
  - **Methanolysis:** PLA + methanol → Methyl lactate.
  - **Pyrolysis:** PLA + heat → Lactide (directly back to monomer).
- **Goal:**
  - True circular economy – PLA bottle → new PLA bottle.





# **Where is PLA Used Today?**

## **(Applications)**

- **Food Packaging**  
Clear cups, salad clamshells, tea bags, coffee pods
- **Food Service**  
Compostable cutlery, straws, plates
- **Medical**  
Dissolving sutures, bone screws, drug delivery
- **3D Printing**  
Most popular filament (easy to print, no odor)
- **Textiles**  
Biodegradable fibers for clothing, wipes, diapers

# Other Developing Bioplastics (Beyond PLA)

- **PHA (Polyhydroxyalkanoates):**
  - Produced by bacteria directly.
  - **Advantage:** Marine biodegradable (breaks down in ocean).
  - **Challenge:** Higher cost than PLA.
- **Starch Blends:**
  - Low cost, used for bags and packaging peanuts.
  - **Disadvantage:** Water-sensitive.
- **Bio-PET / Bio-PE:**
  - Drop-in replacements (same as petro-plastics, but from sugarcane).
  - **Disadvantage:** Not biodegradable.

# Challenges Remaining

- **Not home compostable**  
(requires industrial facility).
- **Poor infrastructure**  
for composting & chemical recycling.
- **Cost premium**  
still 20-50% more expensive than PET/PP.
- **Slow degradation**  
in cold or marine environments.
- **Land use debate**  
food vs. fuel vs. plastic.

**Thank You**